CCIR

Compute Credit Index Research

Benchmark Design Rationale

Design Choices, Rationale, and Implications for Collateral Underwriting

Version:	1.0.0
Date:	March 2026
Status:	Active
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Purpose of This Document

CRI indices are designed to serve as citation-grade reference rates in GPU-backed credit documents — loan covenants, LTV calculations, and liquidation stress scenarios. To serve that function, the index must be constructed with precision: what is measured, what is excluded, and why.

This document records each significant construction decision, the technical rationale behind it, and the practical implication for an underwriter evaluating a borrower's collateral claims. The intent is to give a lender or credit analyst a map: given whatever a borrower is presenting — cluster economics, PCIe hardware, spot pricing, bundled all-in rates — this document explains where that borrower's number sits relative to the reference rate and why.

Each section follows the same structure: the construction choice, the technical rationale, and the underwriter implication.

1. Virtual Machine, Not Container or Kubernetes

CONSTRUCTION CHOICE

CRI indices are computed exclusively from VM-based listings — bare-metal or dedicated VM instances where the GPU is allocated to a single tenant for the duration of the session.

RATIONALE

GPU compute sold through container orchestration layers (Kubernetes, shared namespace pools) does not map to a discrete, allocatable unit of collateral. The CPU, memory, and network resources accompanying a Kubernetes GPU pod are variable, scheduler-managed, and non-exclusive. A VM allocation is a bounded, observable resource unit with a deterministic price per GPU per hour — the correct input for a credit reference rate. Container-based pricing introduces scheduling overhead, multi-tenant interference, and resource ambiguity that would corrupt the index's comparability across sources.

> **UNDERWRITER IMPLICATION**
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> *When a borrower presents utilization or pricing data sourced from a Kubernetes-based platform or a containerized workload scheduler, that data is not directly comparable to CRI. The borrower may be pricing shared or oversubscribed GPU capacity. An underwriter should request the per-GPU per-hour rate on a dedicated VM basis before mapping the borrower's numbers to the reference rate.*

2. Single GPU Unit as the Pricing Atom

CONSTRUCTION CHOICE

Spot index prices are expressed as per-GPU-per-hour rates. All marketplace observations are normalized to a single GPU before entering Silver.

RATIONALE

Multi-GPU listings on GPU marketplaces (pods, instances with 2×, 4×, or 8× GPUs) reflect bundle pricing that may embed block discounts, shared NVLink bandwidth efficiency, or seller inventory management logic. Accepting multi-GPU bundle prices without normalization would introduce a systematic downward bias when large-pod listings dominate the supply snapshot. Single-GPU normalization removes this distortion and establishes a common unit of comparison across all sources and configurations. The per-GPU rate serves as the methodological reference unit for index construction and as the stress floor for credit analysis — it does not assert that GPUs transact individually in a liquidation scenario. In a distressed transaction, the asset is more likely to transfer as a running cluster, a whole facility, or physical servers to a secondary market broker. The per-GPU rate is the baseline against which a lender can test whether a borrower’s projected economics embed a premium that disappears under stress.

> **UNDERWRITER IMPLICATION**

> *A borrower's revenue model built on 8-GPU pod pricing typically embeds a bundle premium, an operator markup, and a utilization assumption that may not survive stress. CRI's per-GPU rate strips all of that out and gives an underwriter a clean baseline: what does one GPU clear at on the open market, noninterruptible, across multiple sources, over a trailing seven-day window? The gap between that number and the borrower's projected per-GPU revenue is a direct read on how much margin is baked into their model and how much cushion disappears if rates compress, utilization falls, or the cluster has to be unwound. Normalizing the borrower's pod economics to a per-GPU rate before comparing to CRI is the correct first step in any collateral stress test.*

3. SXM Interconnect Only in Primary Indices

CONSTRUCTION CHOICE

Primary spot indices (CRI-H100, CRI-H200, CRI-A100, CRI-B200) track SXM-interconnect variants exclusively. PCIe variants are collected and stored but not published as primary indices. CRI-L40S is the exception: the L40S is a PCIe-only GPU with no SXM variant. It is published as a distinct collateral tier and is not treated as a substitute for SXM H100 or A100 collateral.

RATIONALE

SXM and PCIe are not substitutable in practice for the compute-intensive AI training and inference workloads that drive GPU-backed lending. SXM GPUs use NVIDIA's high-bandwidth module-to-module interconnect (NVLink), delivering significantly higher aggregate GPU-to-GPU memory bandwidth than PCIe bus connections. SXM hardware commands a durable premium — typically 30-60% above equivalent PCIe cards — reflecting this architectural difference. Blending SXM and PCIe observations into a single index would introduce a downward bias that understates the collateral value of SXM-equipped infrastructure. Since the majority of GPU-backed credit facilities are underwritten against data center-grade SXM equipment, SXM-pure indices are the more conservative and correct reference.

> **UNDERWRITER IMPLICATION**
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> *A borrower offering PCIe-variant GPUs as collateral — or presenting PCIe rental rates as evidence of market value — is referencing a lower segment of the market. CRI spot indices do not include PCIe observations for H100, H200, A100, or B200. An underwriter comparing a PCIe-heavy H100 portfolio to CRI-H100 (SXM) will understate the haircut required. The correct approach is to apply a PCIe-to-SXM discount before mapping to the index, or to await publication of CCIR's PCIe sub-index, which is collected but not yet published. For L40S collateral specifically, CRI-L40S is the appropriate reference rate — the L40S has no SXM variant, so PCIe is the correct and only form factor for that index.*

4. No Bundled Storage, Networking, or Managed Services

CONSTRUCTION CHOICE

Observations that bundle GPU compute with attached storage, high-bandwidth networking (InfiniBand fabric, private VLAN), or managed services (monitoring, orchestration, SLA guarantees) are excluded from spot index computation or normalized to GPU-only pricing where the bundle components can be isolated.

RATIONALE

Bundled resources are not GPU collateral — they are operating infrastructure. A price that bundles 10TB of NVMe storage with 8×H100s does not convey information about the GPU's standalone market rental rate. In a liquidation or redeployment scenario, storage arrays, networking fabric, and software licenses are separate line items with their own recovery curves. Including bundled prices in a GPU rate index would contaminate the signal with non-GPU assets that may depreciate on entirely different schedules. This is the same principle applied in aircraft EETCs: the airframe appraised value does not include ground support equipment.

> **UNDERWRITER IMPLICATION**
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> *Enterprise borrowers frequently present all-in cluster economics that bundle GPU hours, NVMe storage, InfiniBand networking, and DevOps overhead into a single line-item price. This number is not a GPU price. An underwriter relying on that all-in number to validate CRI coverage is comparing apples to a bundle. The exercise is to strip the non-GPU components from the borrower's pricing model before mapping it to the reference rate. CCIR cluster indices (CLS-MM, CLS-ENT) address multi-GPU commitment pricing — but CLS-MM excludes storage and managed services by design. CLS-ENT (CoreWeave) does include bundled CPU, RAM, storage, and InfiniBand in its per-GPU rate, which is explicitly disclosed in the bundle_note field of every ENT row. An underwriter using CLS-ENT should account for this bundle premium when comparing it to a borrower's GPU-only economics.*

5. Noninterruptible Rate as the Primary Published Rate

CONSTRUCTION CHOICE

CRI indices publish the noninterruptible (on-demand) per-GPU-per-hour rate as the primary rate. Interruptible (spot/preemptible) pricing is collected and stored but not published as the reference rate.

RATIONALE

The correct collateral reference rate for credit purposes is the price at which a qualified counterparty would absorb pledged GPU compute in an orderly redeployment transaction. That counterparty — another operator, a financial buyer, or a receiver — is pricing compute as noninterruptible. No enterprise building a production workload accepts interruptible GPU capacity. Interruptible pricing reflects the shadow price of idle inventory sold to fault-tolerant research and batch workloads — a buyer class not present in a $MM credit liquidation. AVITAS does not appraise aircraft at distressed auction prices; it values them at what a qualified buyer pays in an orderly transaction. The analogous rate for GPU compute is noninterruptible on-demand.

> **UNDERWRITER IMPLICATION**
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> *A borrower citing spot/preemptible GPU prices as evidence of collateral value is presenting the floor of the demand curve — the price at which idle capacity clears, not the price at which deployed productive capacity transfers. The interruptible rate understates recovery value in an orderly liquidation and is therefore the wrong calibration point for a lender's*

> *stress scenario. Conversely, a borrower citing noninterruptible rack rates that exceed CRI noninterruptible should be scrutinized for whether their deployment economics reflect actual market clearing prices or list rates with low utilization.*

6. Marketplace Sources, Not Hyperscaler Rate Cards

CONSTRUCTION CHOICE

CRI spot indices are sourced from GPU cloud marketplaces and specialized GPU cloud providers. Hyperscaler GPU pricing (AWS, Azure, GCP) was evaluated and excluded from the index scope.

RATIONALE

Hyperscaler GPU instances are priced as part of a vertically integrated cloud platform — network egress, identity management, managed Kubernetes, proprietary tooling, and enterprise support are embedded in the price architecture. A borrower's collateral is almost never a hyperscaler GPU instance; it is a dedicated GPU server hosted in a colocation or purpose-built AI data center. The hyperscaler rate card measures what a Fortune 500 enterprise pays for cloud-native GPU access with a full managed stack, not what a standalone GPU compute facility can clear on the open market. Including hyperscaler rates would introduce a persistent upward bias that overstates recoverable collateral value.

> **UNDERWRITER IMPLICATION**
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> *When a borrower references AWS p4d or Azure NDv4 pricing to justify their rate assumptions, they are citing a managed service rate for a product with fundamentally different cost structure and demand drivers. This is not the market CRI measures, and it is not the market in which pledged GPU infrastructure would be liquidated. An underwriter should treat hyperscaler comparisons as indicative of demand-side willingness to pay, not of collateral market clearing rates.*

7. Flat Median Across All Eligible Observations

CONSTRUCTION CHOICE

CRI spot indices are computed as a simple median (percentile_approx) across all Silver-eligible observations in the trailing 7-day window. No source weighting, tier weighting, or outlier trimming is applied.

RATIONALE

A flat median is the correct aggregation method for this pipeline design because the Silver filtering layer — identical apples-to-apples filters applied consistently across all sources — does the cleaning work before aggregation. When Silver filters are correct, a source-weighted median would silently bias the index toward sources with more listings on any given day, rewarding depth over comparability. IQR-based outlier removal was evaluated and rejected: at thin observation counts (which occur for newer GPU models), IQR trimming removes legitimate price signals. Flat median is

auditable, reproducible, and robust to single-source dominance without introducing weighting assumptions that cannot be independently verified.

PROVIDER PRICE DISPERSION

Price dispersion across CRI sources reflects provider wrapper differences, not differences in the underlying GPU asset. Silver comparability filters enforce identical hardware specifications across all sources — same GPU model, same interconnect, same VRAM tier, single-GPU normalized pricing. A listing from a multi-cloud aggregator and a listing from a direct-API rate card provider represent the same physical asset at the same specification. The price difference between them reflects the provider's brand, SLA terms, support relationship, and billing infrastructure — none of which is observable in the GPU itself or relevant to collateral value assessment. An H100 SXM5 rented through a premium provider at $3.34/hr and the same H100 SXM5 rented through a commodity platform at $1.99/hr are identical collateral. The flat median across all Silver-eligible observations therefore produces a rate that reflects where the asset clears across the full market — neither anchored to the premium end nor the commodity floor. For collateral underwriting purposes, this is the correct reference: a lender redeploying distressed GPU collateral will not capture the premium provider rate, but will not necessarily be limited to the commodity floor either. The unweighted median across independently sourced price points is the most defensible estimate of market-clearing rate available without transaction-level data.

Silver Provider Deduplication as Prerequisite

The flat median's integrity depends on Silver delivering one price signal per economic decision-maker per snapshot. A provider's decision to offer compute at a given price is a single market observation, regardless of how many regions, block sizes, or CPU variants that provider deploys at that price. Without this discipline, a provider operating across multiple regions or configuration variants would contribute multiple identical observations to the median pool, inflating its index weight without providing additional price information.

CCIR's Silver layer applies source-specific deduplication rules that normalize each source to one observation per pricing decision before observations enter the flat median: machine_id deduplication for Vast.ai (one physical server = one observation, median taken across block sizes); provider-level deduplication for Shadeform (one upstream cloud provider = one observation, median taken across that provider's available regions); and CPU-variant deduplication for Nebius (one observation per gpu_model per price type per snapshot, eliminating duplicate rows where identical GPU prices are offered across different CPU platform pairings). The result is a median across independent seller prices — not a median inflated by operational deployment choices. This deduplication is what makes the flat median both auditable and resistant to single-source dominance.

The regional median approach for Shadeform providers is specifically suited to collateral valuation: a lender redeploying distressed GPU compute does not know in advance which region the asset will clear in. The median across a provider's available regions is the most defensible estimate of what that provider's capacity would yield across its full deployment footprint — not anchored to a single geography that may or may not be relevant to the actual liquidation transaction. Regional price detail is preserved in Bronze and available for drill-down analysis upon request.

> **UNDERWRITER IMPLICATION**
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> *A borrower or appraiser presenting a weighted average GPU price — weighted by capacity deployed, by source, or by recency — is computing a different statistic than CRI. Depending on the weighting scheme, this can materially move the number above or below the median. An underwriter should understand what weighting methodology underlies any GPU rate claim and whether it would produce a figure above or below the unweighted median. CRI is a conservative, auditable median — not a volume-weighted average that rewards high-listing sources.*

8. Source Independence from the GPU Marketplace

CONSTRUCTION CHOICE

CCIR holds no financial interest in any GPU marketplace, cloud provider, or GPU compute transaction. CCIR is not a market participant, broker, or platform operator.

RATIONALE

The post-LIBOR regulatory environment has established that a benchmark cited in a credit agreement must be structurally independent from the market it measures. A rate produced by a market participant — or calibrated to proprietary, non-auditable data from a single platform — cannot serve as a citation-grade benchmark in a loan covenant or LTV calculation. CCIR's independence is structural, not rhetorical: it derives from the absence of any marketplace relationship, not from a policy statement. This is the same structural requirement that disqualifies lender-controlled metrics and exchange-proprietary rates from being cited as independent benchmarks in debt instruments.

> **UNDERWRITER IMPLICATION**
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> *When evaluating competing GPU rate references — internal models, marketplace-published indices, or proprietary data vendor outputs — an underwriter should ask whether the rate producer has a financial interest in the GPU market it measures. A market data vendor that also operates a trading platform, a marketplace that publishes its own rates, or a lender that derives rates from its own portfolio are all structurally conflicted for covenant citation purposes. CRI is designed to be cited precisely because that conflict does not exist.*

9. Source Evaluation and Exclusion Record

CCIR maintains an internal source evaluation record documenting data providers evaluated and excluded during pipeline development, including the specific construction principles that governed each exclusion decision. This record is retained as part of CCIR's methodology documentation and is available upon written request to parties with executed credit agreements citing CRI rates.

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